

Software Test Plan (STP)

Voice Assistant for Multimedia Controls

Item	Details
Project	Voice Assistant for Multimedia Controls
Team	Team 3
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Source Document	Software Requirements Specification (SRS), Version 1.1

1. Introduction

Purpose: This document defines the software testing plan for the Voice Assistant for Multimedia Controls. It specifies the test objectives, scope, strategy, environment, test cases, traceability, responsibilities, risks, metrics, and acceptance criteria for the ESP32-based voice-controlled multimedia controller.

Scope: Testing covers on-device wake-word detection, voice capture, cloud ASR communication, command recognition and mapping, BLE HID multimedia command transmission, Wi-Fi provisioning, Bluetooth pairing, mute/unmute, factory reset, feedback, diagnostic logging, OTA updates, performance, accuracy, usability, battery operation, modularity, and the five stated security requirements.

Out of scope: Internal media-player implementation, specific streaming-service integration/licensing, multi-language commands, multi-user voice profiles, simultaneous multi-host control, a full-featured mobile app, alternative hardware platforms, fully offline recognition, formal security certification, and long-term production support.

References: Team 3 SRS, Version 1.1

2. Test Items

- Wake Word & Audio Capture module (VAMC-F-001 to F-003)
- Voice Recognition / Cloud ASR module (VAMC-F-004 to F-006)
- Command Execution & Diagnostics module (VAMC-F-007 to F-010, F-014, F-015)
- Connectivity & Pairing module (VAMC-F-011 to F-013)
- System-level integration across wake word → ASR → command mapping → BLE HID → host media action
- Non-functional requirements VAMC-NF-001 to VAMC-NF-005
- Security requirements VAMC-SR-001 to VAMC-SR-005

3. Features to be Tested

Requirement	Feature	Priority	Test Case
VAMC-F-001	Wake-word detection	High	TC-WW-01
VAMC-F-002	Active listening and up to 5-second audio capture	High	TC-WW-02
VAMC-F-003	Physical mute/unmute control	Medium	TC-WW-03
VAMC-F-004	HTTPS streaming to cloud ASR	High	TC-ASR-01
VAMC-F-005	Recognition and mapping to supported commands	High	TC-ASR-02
VAMC-F-006	Unrecognized/ambiguous command handling	High	TC-ASR-03
VAMC-F-007	Seven supported multimedia commands	High	TC-CMD-01
VAMC-F-008	BLE HID command transmission within 2 seconds	High	TC-CMD-02
VAMC-F-009	BLE retry/backoff up to 3 times	Medium	TC-CMD-03
VAMC-F-010	LED/buzzer success feedback	Medium	TC-CMD-04
VAMC-F-011	Wi-Fi provisioning	High	TC-CONN-01
VAMC-F-012	Bluetooth pairing	High	TC-CONN-02
VAMC-F-013	Factory reset	Medium	TC-CONN-03
VAMC-F-014	Diagnostic circular-buffer logging	Medium	TC-DIAG-01
VAMC-F-015	OTA firmware update	Medium	TC-DIAG-02
VAMC-NF-001	End-to-end latency ≤ 3 seconds for 90% of commands	High	TC-PERF-01
VAMC-NF-002	Command recognition accuracy $\geq 90\%$ at SNR ≥ 15 dB	High	TC-ACC-01
VAMC-NF-003	First-time setup average ≤ 5 minutes	Medium	TC-UX-01
VAMC-NF-004	At least 8 hours idle-listening battery life	Medium	TC-POWER-01
VAMC-NF-005	Modular firmware structure	Medium	TC-MAINT-01
VAMC-SR-001 to SR-005	TLS, authenticated BLE, credential protection, audio retention, signed OTA	High/Medium	TC-SEC-01 to 05

4. Features Not to be Tested

- Internal implementation of the host media player (Spotify, VLC, TV OS, etc.).
- Integration or licensing of specific streaming-service APIs.
- Multi-language voice commands.
- Multi-user voice recognition, speaker identification, or personalization.
- Simultaneous control of multiple host devices or multi-room playback.
- Full-featured mobile companion application.
- Raspberry Pi or other alternative hardware platforms.
- Fully offline command recognition.
- Formal security certification beyond the security requirements stated in the SRS.
- Long-term production support, warranty, and manufacturing considerations.

5. Test Approach / Strategy

5.1 Test Levels

- Unit / module testing: Verify each assigned module against its requirements.
- Integration testing: Verify communication between wake-word, ASR, command, connectivity, and diagnostic modules.
- System testing: Verify the complete end-to-end flow from spoken wake word to host media action.
- Acceptance testing: Verify the acceptance criteria and exit criteria defined in the SRS.

5.2 Test Types

- Functional testing
- Integration and system testing
- Regression testing after fixes or module integration
- Performance and latency testing
- Voice recognition accuracy testing
- Usability testing for first-time setup
- Battery/power testing
- Security and privacy validation
- Maintainability/code-structure review

5.3 Cross-Testing Strategy

No member tests their own module. Each module is independently cross-tested by another team member, and all four members jointly perform system-level testing during milestone reviews. Security requirements are reviewed by all members because they cross module boundaries.

Module Under Test	Owner	Cross-Tester	Primary Test Focus
Connectivity & Pairing	Dhanush S S	G S Dheekshith	Wi-Fi provisioning, BLE pairing, factory reset
Wake Word & Audio Capture	G S Dheekshith	Harsh Mannan	Wake-word accuracy/false triggers, mute control
Voice Recognition (ASR)	Harsh Mannan	Chiranthan Shankar	Transcription, command matching, unrecognized commands
Command Execution & Diagnostics	Chiranthan Shankar	Dhanush S S	BLE HID, latency, retry logic, OTA and diagnostics

5.4 Entry Criteria

- ESP32 hardware and microphone available.
- Stable test build available for the module/system under test.
- Required Wi-Fi network and BLE-capable host device available.
- Cloud ASR service is accessible for recognition tests.
- Test utterances and supported command set are prepared.
- Relevant test data and firmware images are available.

5.5 Exit Criteria

- All planned test cases have been executed.
- All high-priority functional requirements are implemented and verified.
- No critical NFR failures remain.
- The RTM shows acceptance status for the applicable requirements.
- Acceptance targets are satisfied, including latency, recognition accuracy, usability, and battery-life targets.
- Security requirements have been reviewed and verified.

6. Security Validation

- Verify all cloud ASR audio traffic uses TLS 1.2 or higher.
- Verify authenticated BLE pairing is used where supported by the host.
- Verify Wi-Fi credentials are not stored in plaintext in flash memory.
- Verify raw audio is not retained after the active recognition session.
- Verify OTA firmware signatures are checked before installation.

7. Test Environment

Category	Required Environment
Hardware	ESP32 development board (e.g., ESP32-WROOM), I2S/analog MEMS microphone, status LED, buzzer, push-button, USB or 2000mAh battery power
Network	2.4 GHz Wi-Fi with stable internet access for cloud ASR
Bluetooth	BLE-capable host device such as smartphone, PC, TV, or Bluetooth speaker supporting BLE HID media keys
Software	ESP32 firmware, cloud Speech-to-Text service, optional Python + MediaPipe PC module
Diagnostics	ESP32 serial/debug interface and local circular diagnostic log
Test Conditions	Typical indoor quiet-to-moderate noise environment; accuracy testing at approximately SNR $\geq$ 15 dB
Test Data	Wake-word utterances, supported command utterances, near-synonyms, ambiguous/unrecognized phrases, Wi-Fi/BLE setup data, signed and unsigned OTA firmware images

8. Test Schedule

The SRS does not define fixed execution dates beyond 05-09-2026. Therefore, the following is a proposed sequence rather than an invented project commitment.

Milestone	Planned Timing	Output
Test planning and case design	After SRS review	Approved STP and test cases
Environment setup	Before module testing	Ready ESP32, Wi-Fi, BLE host and ASR environment
Module testing	After each module reaches T status	Module test results and defects
Integration testing	After module integration	End-to-end results
Performance/security/usability testing	After stable integrated build	NFR and security results
Acceptance testing	Final project review	Acceptance results and sign-off

9. Test Deliverables

- Software Test Plan
- Functional and non-functional test cases
- Security test cases
- Test data / voice utterance set

- Test execution records
- Defect reports
- Requirements Traceability Matrix
- Performance, accuracy, usability, power, and security results
- Final Test Summary Report

10. Roles and Responsibilities

Role	Name	Responsibility
Project Manager & Integration Lead	Dhanush S S	Connectivity/pairing ownership; integration; SRS/RTM/schedule; cross-tests Command Execution & Diagnostics
Audio Module Lead	G S Dheekshith	Wake-word/audio capture and mute control; cross-tests Connectivity & Pairing
Recognition Module Lead	Harsh Mannan	Cloud ASR, transcription and command mapping; cross-tests Audio Module
Output & Maintenance Module Lead	Chiranthan Shankar	BLE HID execution, feedback, diagnostics and OTA; cross-tests Recognition Module
Course Evaluator / Instructor	TBD	Reviews compliance with SRS and acceptance criteria

11. Risks and Mitigation

Risk	Impact	Mitigation
Unstable Wi-Fi / internet	ASR recognition and end-to-end tests may fail	Use a stable test network and record network availability during testing
Cloud ASR availability/rate limits	Recognition tests may be blocked	Schedule ASR tests in controlled batches and record service availability
BLE pairing or host incompatibility	Media command execution cannot be verified	Use a host that supports BLE HID media keys and repeat pairing validation
ESP32 resource limitations	On-device processing may become unreliable	Keep only wake-word processing on-device as specified by the SRS
Background noise	Recognition accuracy may fall below target	Test under the stated SNR condition and separately record degraded-noise observations
Hardware access constraints	Cross-testing may be delayed	Share/rotate available ESP32 hardware according to the team testing plan
Unsigned/tampered OTA image	Security test failure or unsafe update	Verify signature checking and confirm rejection of invalid images

12. Assumptions & Dependencies

- ESP32 is the only hardware platform in scope.
- Wake-word detection runs on-device and full command transcription uses cloud ASR.
- The host supports BLE HID media-key events.
- Stable Wi-Fi is available when cloud recognition is required.
- The optional Python + MediaPipe module is for development/testing and is not required for core operation.
- Testing assumes a typical indoor environment with approximately SNR ≥ 15 dB for the stated accuracy target.
- One configured wake word is used per device.
- The cloud ASR service is accessible for development and demonstration.

13. Suspension & Resumption Criteria

Suspend testing when:

- Required ESP32 hardware or test environment is unavailable.
- The integrated build has a blocking defect preventing meaningful execution.
- Cloud ASR/network dependency is unavailable for tests that require it.
- BLE host compatibility prevents execution of command-transmission tests.

Resume testing when:

- Blocking defects are resolved or a stable build is supplied.
- Hardware, Wi-Fi, BLE, and ASR dependencies are restored.
- Required test data and firmware images are available.

14. Test Case Management & Traceability

The RTM maps each SRS requirement to one or more test cases. Status follows the SRS vocabulary: N = Not Started, P = In Progress, T = Ready for Test, A = Accepted, F = Failed, B = Blocked. Since the SRS identifies the RTM entries as pending actual development progress, all cases below start as N.

TC ID	Req ID	Test Procedure	Expected Result	Status
TC-WW-01	VAMC-F-001	Speak configured wake word at normal volume within 2 m	Listening state is triggered	N
TC-WW-02	VAMC-F-002	Trigger wake word and speak command; observe capture window	Up to 5 seconds of command audio is captured	N
TC-WW-03	VAMC-F-003	Press mute; attempt wake-word activation; unmute	Audio capture is disabled while muted and resumes after unmute	N
TC-ASR-01	VAMC-F-004	Issue supported command over normal Wi-Fi	Audio is sent over HTTPS and transcription round-trip is <3 s under normal Wi-Fi	N
TC-ASR-02	VAMC-F-005	Test exact and near-synonym phrases for all supported actions	Each phrase maps correctly to one of the 7 commands	N
TC-ASR-03	VAMC-F-006	Speak unsupported/ambiguous phrase	Error indication occurs and no unintended command is sent	N
TC-CMD-01	VAMC-F-007	Execute volume up/down, pause, resume, play, next, previous	All 7 commands work independently on the host	N
TC-CMD-02	VAMC-F-008	Measure recognition-to-host execution time	BLE HID event reaches host and executes within 2 s	N
TC-CMD-03	VAMC-F-009	Temporarily make BLE unavailable during command transmission	System retries up to 3 times with backoff, then reports failure	N
TC-CMD-04	VAMC-F-010	Execute a valid command and observe feedback	LED/buzzer confirms success within 1 s	N
TC-CONN-01	VAMC-F-011	Factory-default device: perform guided Wi-Fi setup	Device connects to Wi-Fi within 5 min	N
TC-CONN-02	VAMC-F-012	Pair device with a supported BLE host	Device appears and pairs successfully	N
TC-CONN-03	VAMC-F-013	Long-press factory reset and inspect setup state	Stored Wi-Fi/BLE data is cleared and provisioning restarts	N
TC-DIAG-01	VAMC-F-014	Generate commands/errors and inspect serial/debug log	Last N events are retrievable from the circular buffer	N
TC-DIAG-02	VAMC-F-015	Apply valid signed OTA firmware image	Firmware updates remotely without physical access	N
TC-PERF-01	VAMC-NF-001	Measure wake-word-to-command execution over 50 trials	90th percentile end-to-end latency ≤3 s	N
TC-ACC-01	VAMC-NF-002	Run 100 supported command utterances at SNR ≥15 dB	Recognition accuracy ≥90%	N

TC-UX-01	VAMC-NF-003	Have 5 first-time users perform setup	Average setup time ≤5 min	N
TC-POWER-01	VAMC-NF-004	Run device in idle-listening mode on 2000mAh battery	Continuous operation ≥8 h	N
TC-MAINT-01	VAMC-NF-005	Review module boundaries and replace one module in isolation	Wake-word, ASR and execution modules can be updated independently	N
TC-SEC-01	VAMC-SR-001	Inspect network security during ASR communication	All ASR traffic uses TLS 1.2+	N
TC-SEC-02	VAMC-SR-002	Pair on a host supporting authenticated pairing	Passkey/numeric confirmation is required rather than Just Works	N
TC-SEC-03	VAMC-SR-003	Inspect flash storage after provisioning	Wi-Fi credentials are not stored in plaintext	N
TC-SEC-04	VAMC-SR-004	Inspect memory/flash after command session	Raw audio is not retained after the session	N
TC-SEC-05	VAMC-SR-005	Attempt OTA with unsigned/tampered image	Invalid firmware is rejected before application	N

15. Requirements Traceability Matrix (RTM)

Requirement ID	Requirement / Test Focus	Test Case	Status	Comments
VAMC-F-001	Speak configured wake word at normal volume within 2 m	TC-WW-01	N	Pending execution
VAMC-F-002	Trigger wake word and speak command; observe capture window	TC-WW-02	N	Pending execution
VAMC-F-003	Press mute; attempt wake-word activation; unmute	TC-WW-03	N	Pending execution
VAMC-F-004	Issue supported command over normal Wi-Fi	TC-ASR-01	N	Pending execution
VAMC-F-005	Test exact and near-synonym phrases for all supported actions	TC-ASR-02	N	Pending execution
VAMC-F-006	Speak unsupported/ambiguous phrase	TC-ASR-03	N	Pending execution
VAMC-F-007	Execute volume up/down, pause, resume, play, next, previous	TC-CMD-01	N	Pending execution
VAMC-F-008	Measure recognition-to-host execution time	TC-CMD-02	N	Pending execution
VAMC-F-009	Temporarily make BLE unavailable during command transmission	TC-CMD-03	N	Pending execution
VAMC-F-010	Execute a valid command and observe feedback	TC-CMD-04	N	Pending execution
VAMC-F-011	Factory-default device: perform guided Wi-Fi setup	TC-CONN-01	N	Pending execution
VAMC-F-012	Pair device with a supported BLE host	TC-CONN-02	N	Pending execution
VAMC-F-013	Long-press factory reset and inspect setup state	TC-CONN-03	N	Pending execution
VAMC-F-014	Generate commands/errors and inspect serial/debug log	TC-DIAG-01	N	Pending execution
VAMC-F-015	Apply valid signed OTA firmware image	TC-DIAG-02	N	Pending execution
VAMC-NF-001	Measure wake-word-to-command execution over 50 trials	TC-PERF-01	N	Pending execution
VAMC-NF-002	Run 100 supported command utterances at SNR ≥ 15 dB	TC-ACC-01	N	Pending execution
VAMC-NF-003	Have 5 first-time users perform setup	TC-UX-01	N	Pending execution
VAMC-NF-004	Run device in idle-listening mode on 2000mAh battery	TC-POWER-01	N	Pending execution
VAMC-NF-005	Review module boundaries and replace one module in isolation	TC-MAINT-01	N	Pending execution
VAMC-SR-001	Inspect network security during ASR communication	TC-SEC-01	N	Pending execution
VAMC-SR-002	Pair on a host supporting authenticated pairing	TC-SEC-02	N	Pending execution
VAMC-SR-003	Inspect flash storage after provisioning	TC-SEC-03	N	Pending execution
VAMC-SR-004	Inspect memory/flash after command session	TC-SEC-04	N	Pending execution
VAMC-SR-005	Attempt OTA with unsigned/tampered image	TC-SEC-05	N	Pending execution

16. Test Metrics & Reporting

- Test case execution percentage = executed test cases / planned test cases × 100.
- Pass percentage and fail percentage.
- Requirement coverage = requirements mapped to test cases / total in-scope requirements × 100.
- Defect count and defect severity distribution.
- Defect aging and closure status.
- End-to-end latency statistics, especially 90th percentile over 50 trials.
- Voice command recognition accuracy over 100 utterances.
- Average first-time setup duration over 5 users.
- Battery runtime in idle-listening mode.

Reports: Module test records, cross-testing defect records, integration results, NFR/security results, and the Final Test Summary Report.

17. Approvals

Role	Name	Signature / Date
Team Lead / Student	Dhanush S S	Pending
Course Coordinator / Instructor	TBD	Pending

18. Final Review Checklist

Checklist Item	Status
Project and team details match SRS v1.1	Yes
All 15 functional requirements are represented	Yes
All 5 non-functional requirements are represented	Yes
All 5 security requirements are represented	Yes
Test cases map to the SRS requirement IDs	Yes
Cross-testing assignments follow the team's no-self-testing rule	Yes
Actual execution status	Pending project execution
Instructor/team approval signatures	Pending